Cambridge (CIE) IGCSE Additional Maths

Your notes

Circular Measure

Contents

Radian Measure

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Radian Measure

What are radians?

- Radians are an alternative to degrees for measuring angles
- 1 radian is the angle in a **sector** of radius 1 and arc length 1
  - A circle with radius 1 is called a **unit circle**
- Radians are normally quoted in terms of π
  - 2π radians = 360°
  - π radians = 180°
- The symbol for radians is ^{c} but it is more usual to see **rad**
  - Often, when π is involved, no symbol is given as it is obvious it is in radians
  - Whilst it is okay to omit the symbol for radians, you should never omit the symbol for degrees
- In the exam you should use radians unless otherwise indicated

Definition of a radian using a unit circle

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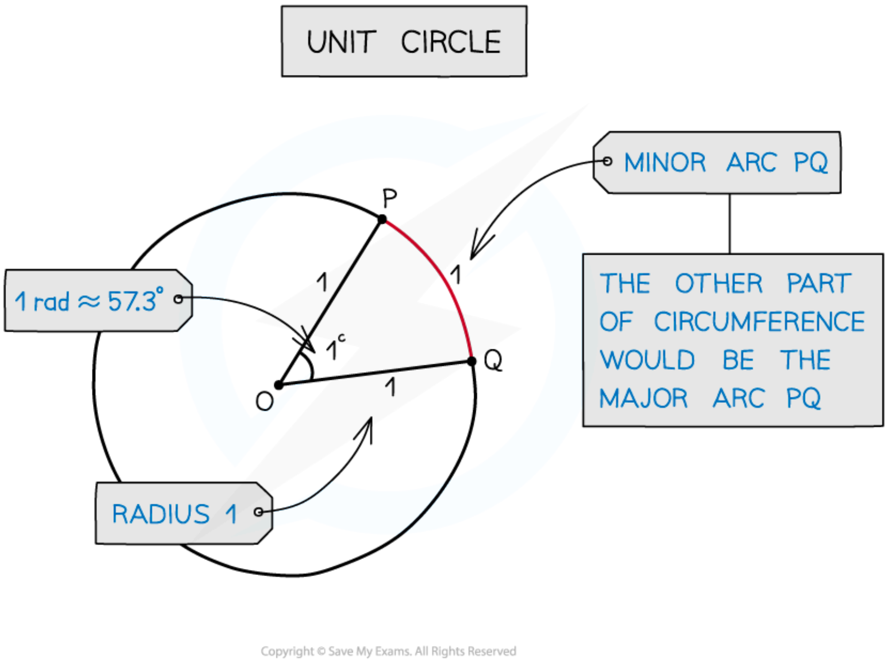

How do I convert between radians and degrees?

- Use $\pi^c = 180°$ to convert between radians and degrees
  - To convert from radians to degrees multiply by $\frac{180}{\pi}$
  - To convert from degrees to radians multiply by $\frac{\pi}{180}$
- Some of the common conversions are:
  - $2\pi^c = 360°$
  - $\pi^c = 180°$
  - $\frac{\pi}{2}^c = 90°$
  - $\frac{\pi}{3}^c = 60°$

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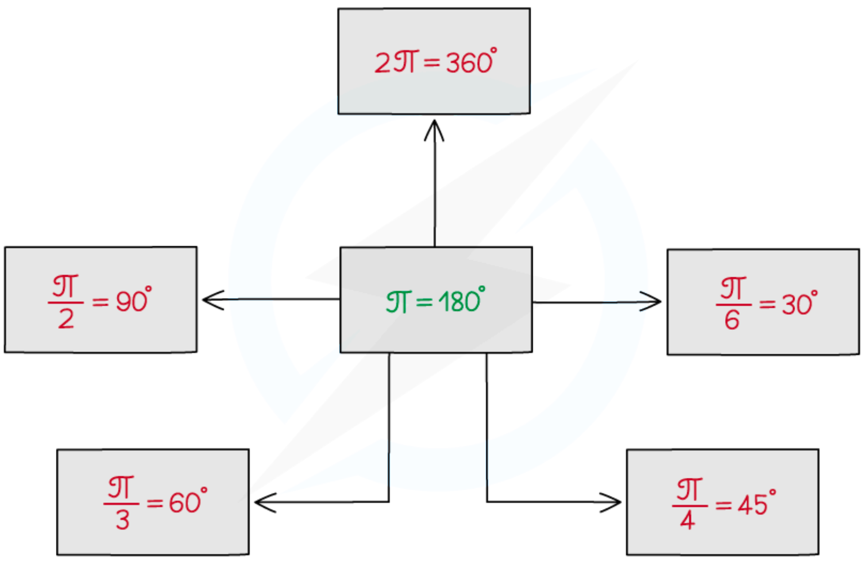
- $\dfrac{\pi}{4}^{c} = 45°$
- $\dfrac{\pi}{6}^{c} = 30°$
- It is a good idea to remember some of these and use them to work out other conversions



Examiner Tips and Tricks

- Sometimes an exam question will specify whether you should be using degrees or radians and sometimes it will not, if it doesn't it is expected that you will work in radians
- If the question involves π then working in radians is useful as there will likely be opportunities where you can cancel out π
- Make sure that your calculator is in the correct mode for the type of angle you are working with

Worked Example

a) Convert 43.8° to radians.

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Divide by 180°.

$$\frac{43.8°}{180°} = \frac{73}{300}$$

Multiply by π radians.

$$\frac{73}{300} \times \pi$$

$$\mathbf{43.8° = \frac{73}{300}\pi = 0.764\ (3sf)}$$

b) Convert $\frac{5\pi}{4}$ to degrees.

Divide by π radians.

$$\frac{5\pi}{4} \div \pi = \frac{5}{4}$$

Multiply by 180°.

$$\frac{5}{4} \times 180°$$

$$\mathbf{\frac{5\pi}{4}\ \text{radians} = 225°}$$

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Arcs & Sectors

Length of an Arc

What is an arc?

- An arc is a part of the **circumference** of a circle
  - It is easiest to think of it as the crust of a single slice of pizza
- The length of an arc depends of the size of the angle at the centre of the circle
- If the angle at the centre is **less than 180°** then the arc is known as a **minor arc**
  - This could be considered as the crust of a single slice of pizza
- If the angle at the centre is **more than 180°** then the arc is known as a **major arc**
  - This could be considered as the crust of the remaining pizza after a slice has been taken away

How do I find the length of an arc?

- The length of an arc is simply a fraction of the circumference of a circle
  - The fraction can be found by dividing the angle at the centre by 360°
- The formula for the length, l, of an arc is

$$l = \frac{\theta}{360} \times 2\pi r$$

- Where θ is the angle measured in degrees
  - r is the radius

How do I use radians to find the length of an arc?

- As the radian measure for a **full turn** is 2π, the fraction of the circle becomes $\frac{\theta}{2\pi}$
- Working in radians, the formula for the length of an arc will become

$$l = \frac{\theta}{2\pi} \times 2\pi r$$

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- Simplifying, the formula for the length, l, of an arc is

$$l = r\theta$$

- - θ is the angle measured in **radians**
  - r is the radius

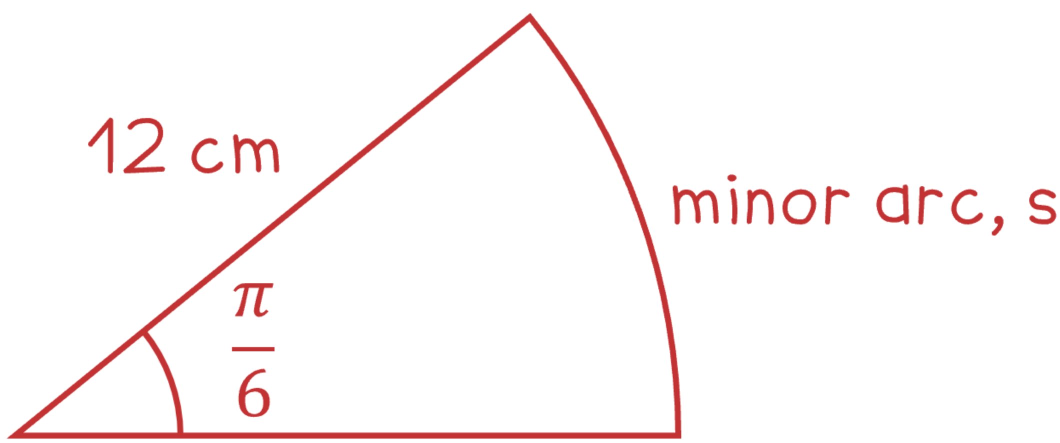
Worked Example

A circular pizza has had a slice cut from it, the angle of the slice that was cut was $\frac{\pi}{6}$ rad.

The radius of the pizza is 12 cm. Find

i) the length of the outside crust of the slice of pizza (the minor arc),

A simple diagram will help



The formula for the length of an arc, where the angle is in radians is $s = r\theta$

$$s = 12 \times \frac{\pi}{6}$$

2π cm

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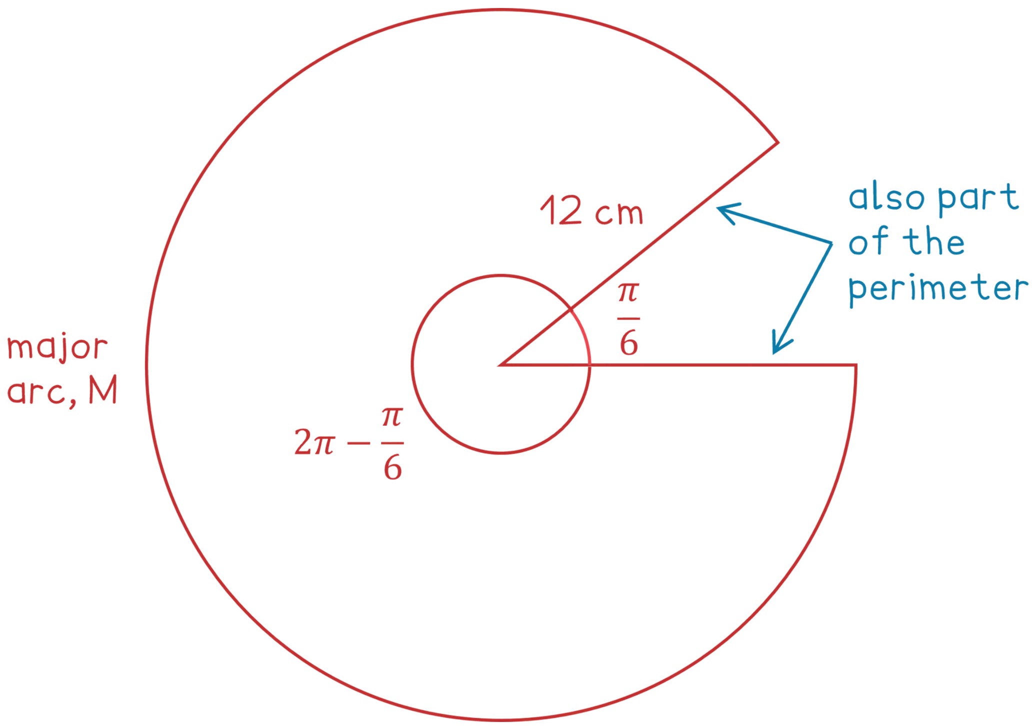
ii) the perimeter of the remaining pizza.

A diagram will help consider where the perimeter is



Find the angle for the major arc, by subtracting from the angle in a full circle

$$2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$$

Use the formula for the length of an arc, $s = r\theta$, to find the curved length of the perimeter, the major arc M

$$M = 12 \times \frac{11\pi}{6} = 22\pi$$

As we are finding the perimeter of the whole shape, we need to add on the two straight lengths formed by the slice which has been cut out

$$22\pi + 12 + 12$$

$$\mathbf{22\pi + 24\ cm}$$

Unless asked to otherwise, it is best to give answers in an exact form

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Area of a Sector

What is a sector?

- A sector is a part of a circle enclosed by two radii (radiuses) and an arc
  - It is easier to think of this as the shape of a single slice of pizza
- The area of a sector depends of the size of the angle at the centre of the sector
- If the angle at the centre is **less than 180°** then the sector is known as a **minor sector**
  - This could be considered as the shape of a single slice of pizza
- If the angle at the centre is **more than 180°** then the sector is known as a **major sector**
  - This could be considered as the shape of the remaining pizza after a slice has been taken away

How do I find the area of a sector?

- The area of a sector is simply a fraction of the area of the whole circle
  - The fraction can be found by dividing the angle at the centre by 360°
- The formula for the area, A, of a sector is

$$A = \frac{\theta}{360} \times \pi r^2$$

- Where θ is the angle measured in degrees
  - r is the radius

How do I use radians to find the area of a sector?

- As the radian measure for a **full turn (360°)** is 2π, the fraction of the circle becomes $\frac{\theta}{2\pi}$
- Working in radians, the formula for the area of a sector will become

$$A = \frac{\theta}{2\pi} \times \pi r^2$$

- Simplifying, the formula for the area, A, of a sector is

$$A = \frac{1}{2} r^2 \theta$$

- θ is the angle measured in **radians**
  - r is the radius

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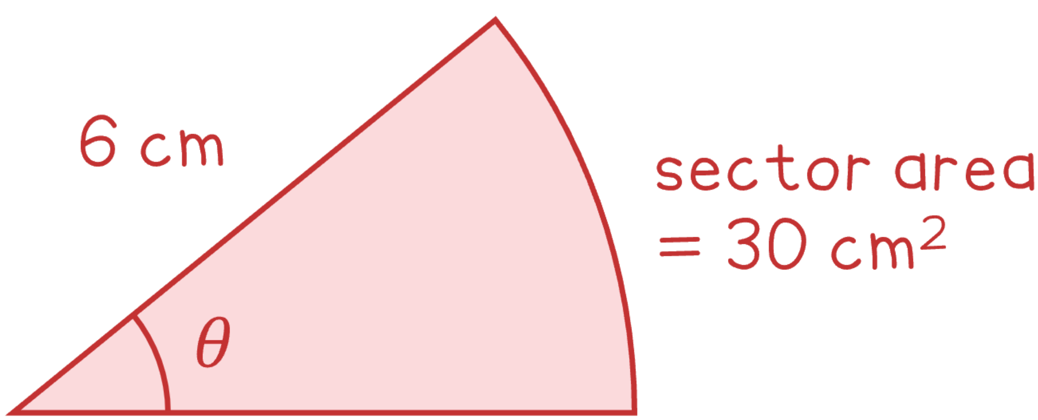
Examiner Tips and Tricks

- These formulae are **not** given to you - you need to remember them!
- Make sure that you read the question carefully to determine exactly what you need to calculate
  - the arc length or area of a sector,
  - the perimeter or area of a compound shape,
  - or something else that incorporates the arc length or area!

Worked Example

A sector of radius 6 cm has an area of 30 cm^2. Find the angle at the centre of the sector in radians.

Draw a diagram to help



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Use the formula for the area of a sector, where the angle is in radians; $A = \frac{1}{2}r^2\theta$

$$30 = \frac{1}{2} \times 6^2 \times \theta$$

Solve for θ

$$\theta = \frac{30 \times 2}{6^2}$$

$$\mathbf{\frac{5}{3}}\ \textbf{radians}$$